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PS 01: AI-BASED NATIONAL CYBER THREAT EARLY WARNING SYSTEM

PROBLEM STATEMENT

Cyber threats are evolving rapidly, with attackers exploiting newly discovered vulnerabilities, launching phishing campaigns, distributing malware, and targeting digital infrastructure. Security teams often receive threat information from multiple sources, making it difficult to identify emerging threats, correlate related incidents, and determine which systems require immediate attention.

Traditional security approaches often react after suspicious activity has already occurred. There is a need for an intelligent, proactive system that can analyse available cyber threat intelligence, identify emerging attack patterns, and generate early warnings before threats develop into major security incidents.

CHALLENGE

Develop an AI-powered Cyber Threat Early Warning System that collects and analyses threat intelligence from multiple sources to identify suspicious patterns, emerging attack campaigns, and potentially vulnerable systems.

The system should transform raw threat information into actionable intelligence, helping security analysts understand what is happening, which assets may be at risk, and what preventive measures should be prioritised.

EXPECTED SOLUTION

The proposed solution should demonstrate how AI can help security teams move from reactive incident response towards proactive threat identification and prevention.

A working prototype could include a dashboard showing incoming threat intelligence, an AI-generated threat summary, a prioritised risk score, and an early warning explaining why a particular threat deserves attention.

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KEY OBJECTIVES

- Threat Intelligence Aggregation: Collect and consolidate threat information from trusted feeds, public advisories, vulnerability databases, and simulated security logs.
- AI-Based Threat Analysis: Apply machine learning or natural language processing to identify suspicious indicators, recurring attack patterns, and relationships between reported threats.
- Early Warning Generation: Identify potentially emerging threats and generate timely alerts based on threat severity, confidence, and relevance.
- Risk Prioritisation: Rank threats according to factors such as exploitability, asset criticality, potential impact, and available evidence.
- Actionable Recommendations: Provide suggested mitigation measures, such as patching vulnerable systems, blocking known malicious indicators, or investigating suspicious activity.
- Threat Intelligence Dashboard: Display active threats, affected technologies

Example use case: A newly published vulnerability advisory is correlated with simulated asset inventory data showing that an organisation uses the affected software. The system identifies the potential exposure, assigns a risk level, and generates an alert recommending investigation and remediation.

EXPECTED DELIVERABLES

- A functional prototype of the AI-based early warning system.
- A dashboard displaying detected threats, risk levels, and alerts.
- A demonstration using public threat intelligence or a controlled, simulated dataset.
- An explanation of the AI model, threat scoring approach, and proposed mitigation workflow.